

Practice Problems List-4 (If-else)

1. Find the largest one among three integers.
2. Write a program that takes the cost price and selling price of an item as input and calculates the profit or loss. If there is a profit, print the amount of profit. Else if there's loss, print the amount of loss. Otherwise print **No Profit or Loss**.

Input: 100 120

Output: Profit of 20

Input: 150 80

Output: Loss of 70

3. Write a program that takes two integers and a character (+, -, *, /, %) as input. Based on the character, perform the corresponding arithmetic operation between the two numbers and print the result. If the input operator is invalid, print an error message.

Input: 5 3 +

Output: 8

4. Take the marks (integer) of a student as input and check if he failed (obtained less than 40) in the subject or not.
5. Take a year (integer) as input and check Whether the entered year is Leap Year or not.
(If the year is divisible by 100, it MUST BE divisible by 400 to be a leap year. All other years are leap years if they are simply divisible by 4)

6. Take marks (integer) of a student as input and print his grade as per the following chart:

Marks	Grade
0-39	F
40-59	C
60-69	B
70-79	A
80-100	A+
Below zero or above 100	Invalid

7. Write a C program that takes a month number (1-12) and a year as input and prints the number of days in the month. Handle leap years for February.

Input: 2 2024

Output: 29

8. Write a C program to check whether a given 5-digit integer is a palindrome or not.

Input: 15251

Output: Palindrome

Input: 15315

Output: Not Palindrome

9. Take a 5-digit integer and input and print the number of odd digits in it.

Input: 15251

Output: 4

10. Take a temperature (a decimal number) and a character signifying the scale as input. If the character is **C**, that means the temperature is in Celsius. You have to convert it to Fahrenheit. If the character is **F**, you have to convert it to Celsius.

Input: **36.7 C**

Output: **98.06**

11. Take the age of a person (integer) as input and check if he is eligible to vote or not (at least 18 years old or not).
12. Take input the sizes of three sides of a triangle and check whether the triangle is equilateral, scalene, or isosceles.
(equilateral means all sides are equal, isosceles means two sides are equal, scalene means all three sides are of different size)
13. Take input the measurement of three sides (integer) of a triangle and check whether the triangle is valid or not
(sum of any two sides must be greater than the third side)
14. Take input the measurement of three angles (integer) of a triangle and check whether the triangle is valid or not (sum of three angles must be 180)
15. Take two integer numbers (x,y) as input as if they represent a point in the XY coordinate plane, and print which quadrant they belong to.
(For example- the point (1,5) belongs in the first quadrant whereas (-1,2) belongs in the second.)
16. Build a simple calculator. Take input two numbers and a character. If the character is '+', add the two numbers. Similarly do the relevant operations for '-', '*', '/' etc. See the sample input and output below:

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Enter the first number: 10
Enter the second number: 5.5
Enter the operation: *
5.50 * 0.00 = 0.00
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17. Write a program that calculates the Body Mass Index (BMI) based on the input weight (in kilograms) and height (in meters).

Use the formula: $BMI = weight/height^2$

Based on the calculated BMI, print the category:

- Underweight: BMI < 18.5
- Normal weight: BMI 18.5–24.9
- Overweight: BMI 25–29.9
- Obese: BMI $\geq$ 30

Input: 70 1.75

Output: BMI = 22.86 (Normal weight)

18. Write a C program to calculate the electricity bill based on the following criteria:

- For the first 100 units: BDT 10.00 per unit
- For the next 200 units: BDT 15.00 per unit
- For units above 300: BDT 20.00 per unit
- Add a surcharge of 10% on the total bill if it exceeds BDT 2000.00.